Counterfactual Training

Teaching Models Plausible and Actionable Explanations

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Intro

A Toy Problem

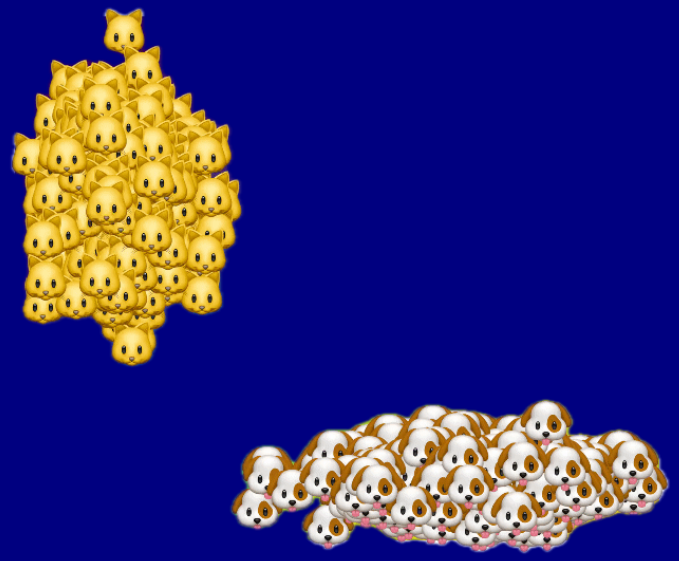

Figure 1: Cats and dogs in two dimensions.

Traversing the Parameter Space

Model Training

Objective:

$$\min_{\theta}\{\text{yloss}(M_\theta(\mathbf{x}), \mathbf{y})\}$$

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Solution:

$$\theta_{t+1} = \theta_t - \nabla_\theta\{\text{yloss}(M_\theta(\mathbf{x}), \mathbf{y})\}$$
$$\theta^* = \theta_T$$

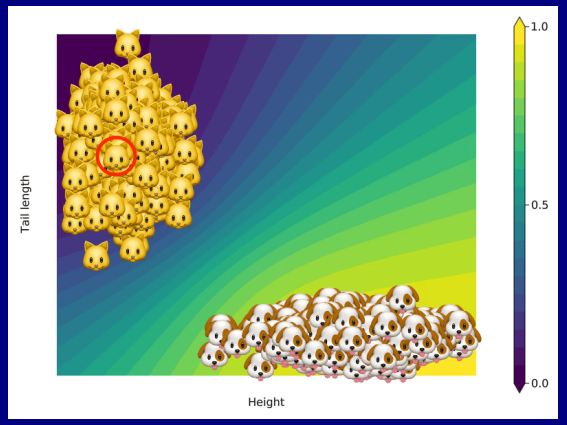

Figure 2: Fitted model. Contour shows predicted probability $y =$.

Traversing the Feature Space

Counterfactual Search

Objective:

$$\min_{\mathbf{x}}\{\text{yloss}(M_{\theta^*}(\mathbf{x}), \mathbf{y}^+) + \lambda\text{reg}\}$$

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Counterfactual Search

Objective:

$$\min_{\mathbf{x}}\{\text{yloss}(M_{\theta^*}(\mathbf{x}), \mathbf{y}^+) + \lambda\text{reg}\}$$

Solution:

$$\mathbf{x}_{t+1} = \mathbf{x}_t - \nabla_\theta\{\text{yloss}(M_{\theta^*}(\mathbf{x}), \mathbf{y}^+)\}$$
$$\mathbf{x}^* = \mathbf{x}_T$$

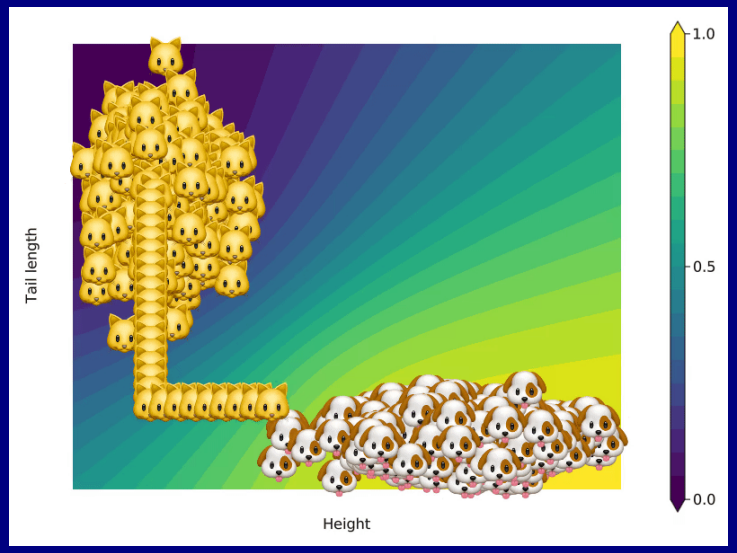

Figure 3: Counterfactual explanation for what it takes to be a dog.

Plausibility at all cost?

All of these counterfactuals are valid explanations for the model's prediction.

Pick your poison ...

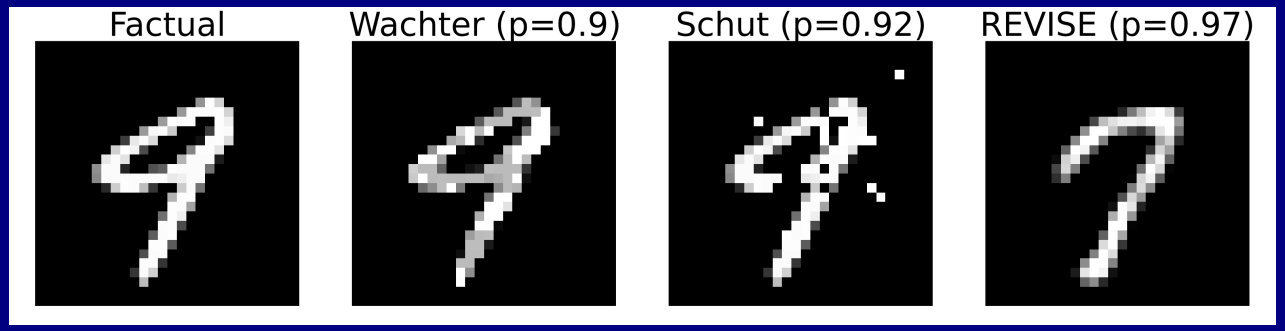

Figure 4: Turning a 9 into a 7: Counterfactual explanations for an image classifier produced using *Wachter* (Wachter, Mittelstadt, and Russell 2017), *Schut* (Schut et al. 2021) and *REVISE* (Joshi et al. 2019).

Faithful First, Plausible Second

Counterfactuals as plausible as the model permits[1].

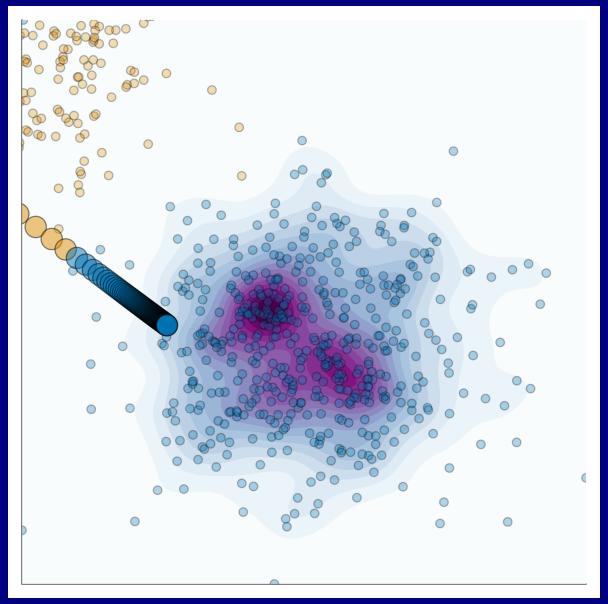

Figure 5: KDE for training data.

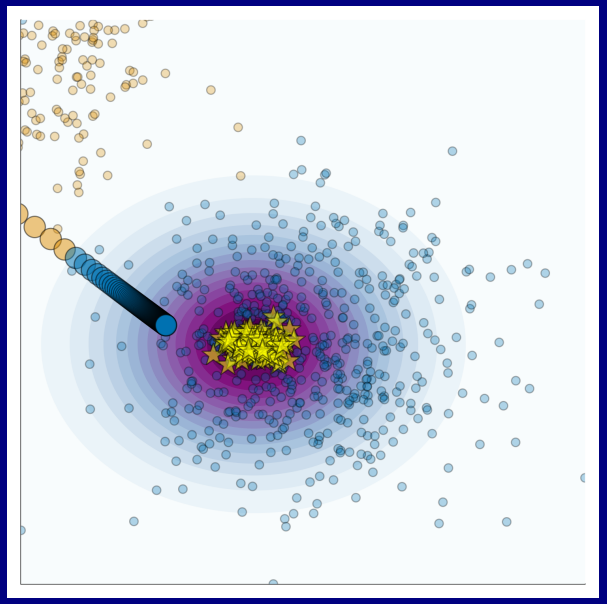

Figure 6: KDE for model posterior.

[1] Altmeyer et al. (2023) @ AAAI 2024. [blog]

Faithful Counterfactuals

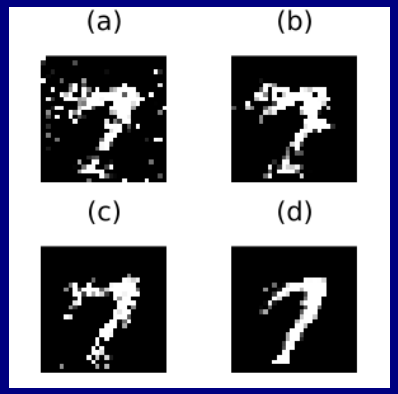

Figure 7: Turning a 9 into a 7. *ECCCo* applied to MLP (a), Ensemble (b), JEM (c), JEM Ensemble (d).

🔑 **Insight**: faithfulness facilitates

- model quality checks (Figure 7).
- state-of-the-art plausibility (Figure 8).

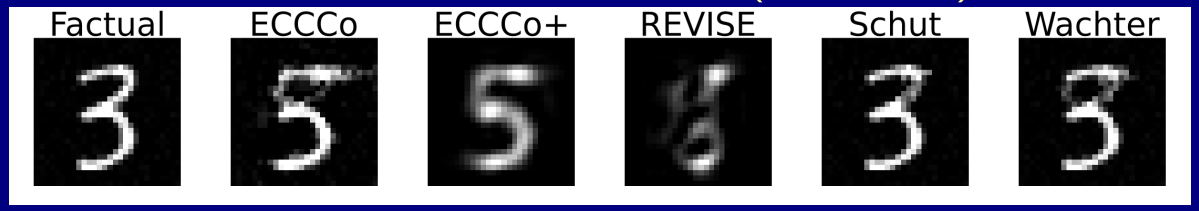

Figure 8: Results for different generators (from 3 to 5).

New Opportunities

Counterfactual Training: Method

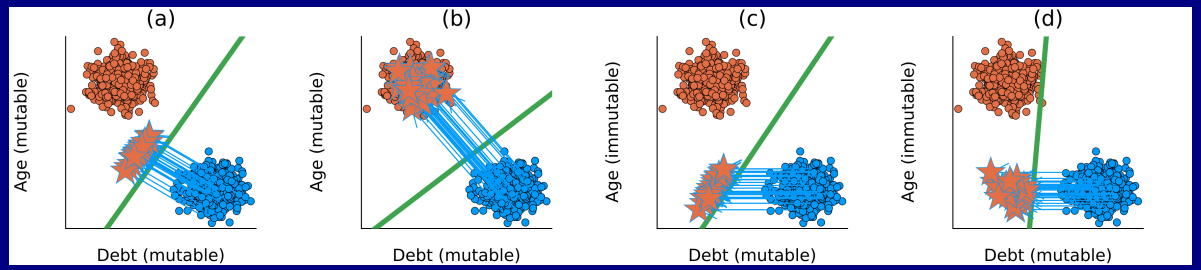

Figure 9: (a) conventional training, all mutable; (b) CT, all mutable; (c) conventional, *age* immutable; (d) CT, *age* immutable.

1. Contrast faithful CE with data.
2. Enforce actionability constraints.
3. Bonus: use nascent CE as AE.

Insight: We can hold models accountable for plausible explanations[2].

Counterfactual Training: Results

- Models trained with CT learn more plausible and (provably) actionable explanations.
- Predictive performance does not suffer, robust performance improves.

Figure 10: *Plausibility*: BL (top row) vs CT using the *ECCCo* generator (bottom row) counterfactuals for a randomly selected factual from class "0" (in blue). CT produces more plausible counterfactuals than BL.

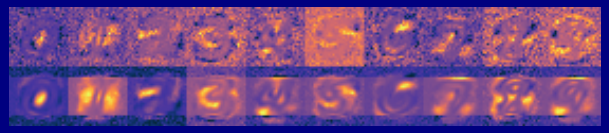

Figure 11: *Actionability*: Sample visual explanations (integrated gradients) for the *MNIST* dataset. Mutability constraints are imposed on the five top and bottom rows of pixels. CT (bottom) is less sensitive to protected features.

References

Altmeyer, Patrick, Aleksander Buszydlik, Arie van Deursen, and Cynthia C. S. Liem. 2026. "Counterfactual Training: Teaching Models Plausible and Actionable Explanations." In *2026 IEEE Conference on Secure and Trustworthy Machine Learning (SaTML)*. IEEE. upcoming.

Altmeyer, Patrick, Mojtaba Farmanbar, Arie van Deursen, and Cynthia C. S. Liem. 2023. "Faithful Model Explanations Through Energy-Constrained Conformal Counterfactuals." https://arxiv.org/abs/2312.10648.

Joshi, Shalmali, Oluwasanmi Koyejo, Warut Vijitbenjaronk, Been Kim, and Joydeep Ghosh. 2019. "Towards Realistic Individual Recourse and Actionable Explanations in Black-Box Decision Making Systems." https://arxiv.org/abs/1907.09615.

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